

FRAN SÁNCHEZ RODRIGO

[Unreal Engine] Tools, Engine & Optimization Engineer

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EXPERIENCE

Unreal Tools & Engine Programmer Unreal Console Porting Specialist

at Catness Game Studios

Dec 2021 - Mar 2025

- Game porting using Unreal Engine 4 and 5 to console platforms:
 - [Xuan Yuan Sword 7](#) (Switch)
 - [Metamorphosis VR](#) (Oculus Quest 2)
 - [American Arcadia](#) (Switch, PS4)
- Plugin and tool development to extend engine functionality:
 - Runtime-applicable custom device profiles.
 - Project analysis and statistics tools.
 - Custom Blueprint nodes.
 - Context menu extension framework to create custom tasks.
- Plugin and tool development to streamline console porting:
 - Xbox user management and controller handling.
 - PS5 Activities management.
 - Optimization tools (automatic MaxDrawDistance, tick management, mesh instancing).
 - Automatic platform-based asset replacement.
- General engine modifications and fixes:
 - Indexed Actor tags optimization.
 - Automatic button hover behavior for consoles.
 - Mobile Software Occlusion fixes.
 - Improved MinLOD behavior for consoles.
 - Custom build deployment with folder structure and visible build hashes.

Quality Assurance Technician

at Lollipop Robot

Feb 2016 - Sep 2020

- Playtesting and Compliance testing for PC and consoles (Xbox One, PS4, Switch).
- Experience using popular bug trackers like Jira, Mantis, Bitbucket, or GitHub.

EDUCATION

Master's Degree in Video Game Programming

2020 - 2021

U-Tad (Madrid)

Graduate (Bachelor's Degree) in Video game Design and Development

2012 - 2016

Universitat Jaume I (Castellón)

SKILLS AND COMPETENCIES

- Advanced proficiency in game programming with Unreal Engine (C++).
- Intermediate level of game development with Unity.
- High-level experience with version control systems (Git, Perforce).
- Fast adaptability to new codebases.
- Ability to work with low-level SDKs and systems.
- Experience with multiple programming and scripting languages: Java, C#, Python, Lua.
- Basic knowledge of CMake for C++ development.

LANGUAGES

- English: Professional working proficiency.
- Spanish: Native or bilingual proficiency.
- Catalan: Native or bilingual proficiency.

ACADEMIC PROJECTS

Ronin-Hood Sisters

Master's Degree in Video Game Programming

Tactical RPG made with Unreal Engine 4 (C++), inspired by magical girls with elemental powers and destructive abilities.

I worked on programming the game world, a custom map creation tool, the fog of war system, the save system, and the logic behind multiple UI elements, including the costume selection screen, the inventory, and the combat HUD.

- [Detailed explanation of all my work in the game](#)

Medieval Floor

Master's Degree in Video Game Programming

Action game made with Unreal Engine 4 (C++). Local cooperative game for up to 4 players, where you clear waves of increasing difficulty.

I worked on programming the gameplay flow, including enemy pool generation, wave-based spawning, and saving high scores to disk.

- [Detailed explanation of all my work in the game](#)
- [Check the source code](#)

Unity Zombie Killer

Master's Degree in Video Game Programming

First-person zombie survival game where players must complete three different quests to win.

Developed with Unity (C#) in 5 days.

I worked on programming the weapon systems, pickable items, zombie AI, and the quest system.

- [Detailed explanation of all my work in the game](#)
- [Check the source code](#)

OTHER PROJECTS

Unused Node Cleaner

"Unused Node Cleaner" is an Unreal Engine plugin that allows you to clean Blueprints by removing unused nodes left behind in your graphs, reducing visual noise and improving readability.

- [Fab.com download page](#)

Quantum fields

Entry for the 2023 Epic MegaJam. A portal-like game where you can open portals to see the past.

I implemented an asynchronous dynamic mesh cutting system.

- [Detailed explanation of all my work in the game](#)

Urkaz Moon Tools

A "Minecraft: Java Edition" mod that adds two items to check the moon phase. Its main purpose is to be used in conjunction with other mods. Written in Java.

- [Github repository](#)